

Bloch

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Chapter 1

Construction of the Real Numbers

Definition 1.1. Let S be a set. A **binary operation** on S is a function $S \times S \rightarrow S$. A **unary operation** on S is a function $S \rightarrow S$.

Let S be a set and let $*$: $S \times S \rightarrow S$ be a binary operation. If $x, y \in S$, the correct way to write the result of the operation to the pair $\langle x, y \rangle$ would be $*(\langle x, y \rangle)$. However since this is a bit cumbersome one may write $x * y$ instead. Similarly if $\neg$: $S \rightarrow S$ is a unary operation and $x \in S$ one may choose to write $\neg x$ over $\neg(x)$.

1.1 Axioms for the Natural numbers

1.1.1 Peano Postulates

Axiom: Peano Postulates

There exists a set $\mathbb{N}$ with an element $1 \in \mathbb{N}$ and a function $s : \mathbb{N} \rightarrow \mathbb{N}$ that satisfy the following three properties.

- There is no $n \in \mathbb{N}$ such that $s(n) = 1$.
- The function s is injective.
- Let $G \subseteq \mathbb{N}$ be a set. Suppose that $1 \in G$, and that if $g \in G$ then $s(g) \in G$. Then $G = \mathbb{N}$.

It will be shown that the set $\mathbb{N}$ is unique. We can therefore give it a name.

Definition 1.2. The set of **natural numbers** denoted $\mathbb{N}$, is the set the existence of which is given in the Peano Postulates.

1.1.2 Consequences of Peano Postulates

Lemma 1.3. *Let $a \in \mathbb{N}$. Suppose that $a \neq 1$. Then there is a unique $b \in \mathbb{N}$ such that $a = s(b)$. (where existence of s and $\mathbb{N}$ are outlined by the peano axioms)*

Proof. First we prove existence. Let

$$G = \{1\} \cup \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$$

(which exists by the subset, pairing, and union axioms) We will prove that $G = \mathbb{N}$. To do this we need to show $1 \in G$, $G \subseteq \mathbb{N}$ and $\forall g \in G(s(g) \in G)$. The first two are straightforward. For the third suppose $g \in G$; we have two cases. First suppose $g \in \{1\}$; then $g = 1 \in \mathbb{N}$. Since $s : \mathbb{N} \rightarrow \mathbb{N}$, we then have $s(1) \in \mathbb{N}$. So by definition $s(1) \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ and $s(g) = s(1) \in G$. Now suppose $g \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$; then by definition $g \in \mathbb{N}$ and $s(g) \in \mathbb{N}$ exists. By definition $s(g) \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ so $s(g) \in G$. By the third peano postulate we then can assert that $G = \mathbb{N}$. So $a \in \mathbb{N} = G$. Since $a \neq 1$ we must then have $a \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ and by definition, $\exists b \in \mathbb{N}(s(b) = a)$.

Now for uniqueness. Let m and n be arbitrary elements of $\mathbb{N}$ such that $a = s(m)$ and $a = s(n)$. Then $\langle m, a \rangle \in s$ and $\langle n, a \rangle \in s$. Since s is injective we must have $n = m$. $\square$

Theorem 1.4. *Let H be a set, Let $e \in H$ and let $k : H \rightarrow H$ be a function. Then there is a unique function $f : \mathbb{N} \rightarrow H$ such that $f(1) = e$ and $f \circ s = k \circ f$.*

Proof. Letting H, e , and k be arbitrary such that $e \in H$ and $k : H \rightarrow H$. We will show existence and uniqueness separately. We start with uniqueness.

Let g and h be arbitrary such that $g : \mathbb{N} \rightarrow H$, $g(1) = e$, and $g \circ s = k \circ g$ and $h : \mathbb{N} \rightarrow H$, $h(1) = e$, and $h \circ s = k \circ h$. We will first define V as the unique set (guaranteed by the subset and extensionality axioms) such that

$$\forall a(a \in V \leftrightarrow a \in \mathbb{N} \wedge g(a) = h(a))$$

and show that $V = \mathbb{N}$. By peano's postulates, we need to show $V \subseteq \mathbb{N}$, $1 \in V$, and $\forall n \in V(s(n) \in V)$. The first is straightforward. For the second requirement, we know by peano's postulates that $1 \in \mathbb{N}$, so by our assumptions $g(1)$ and $h(1)$ exist and $g(1) = e = h(1)$. So $1 \in V$.

For the third requirement, let n be arbitrary and suppose that $n \in V$. Then by definition $n \in \mathbb{N}$ and $g(n) = h(n)$. Since $n \in \mathbb{N} = \text{dom}(h \circ s)$, we have, by our assumptions,

$$h(s(n)) = h \circ s(n) = k \circ h(n) = k(h(n)) = k(g(n)) = k \circ g(n) = g \circ s(n) = g(s(n))$$

we know by definition $\text{ran}(s) \subseteq \mathbb{N}$, so $s(n) \in \mathbb{N}$ and again by definition $s(n) \in V$. We can conclude that $V = \mathbb{N}$.

Suppose $\langle x, y \rangle \in g$, then $x \in \text{dom } g = \mathbb{N} = V$. So by definition $y = g(x) = h(x)$, and $\langle x, y \rangle \in h$. Suppose $\langle x, y \rangle \in h$, then $x \in \text{dom } h = \mathbb{N} = V$. So by definition $y = h(x) = g(x)$, and $\langle x, y \rangle \in g$.

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Now for existence. We need to come up with a function from $\mathbb{N}$ into H such that $f(1) = e$ and $f \circ s = k \circ f$. We first define C , where

$$\forall X (X \in C \leftrightarrow X \subseteq \mathbb{N} \times H \wedge \langle 1, e \rangle \in X \wedge \forall n \forall y (\langle n, y \rangle \in X \rightarrow \langle s(n), k(y) \rangle \in X))$$

(guaranteed by the subset and extensionality axioms.) We propose that $f = \bigcap C$ is the desired function. To ensure its existence, we need to show that C is nonempty. We will show that $\mathbb{N} \times H \in C$. For the first requirement, clearly $\mathbb{N} \times H \subseteq \mathbb{N} \times H$. Next, since $1 \in \mathbb{N}$ and $e \in H$ then $\langle 1, e \rangle \in \mathbb{N} \times H$. Finally, let n and y be arbitrary such that $\langle n, y \rangle \in \mathbb{N} \times H$. Since $s : \mathbb{N} \rightarrow \mathbb{N}$ and $k : H \rightarrow H$, we must then have $s(n) \in \mathbb{N}$ and $k(y) \in H$. So $\langle s(n), k(y) \rangle \in \mathbb{N} \times H$. $f = \bigcap C$ is therefore a valid candidate.

Before we prove the functionality of f , we first state a few facts about f . First we note that $f \in C$. Let arbitrary $x \in \bigcap C$. Since C is nonempty there exists $X \in C$, and by definition $x \in X \subseteq \mathbb{N} \times H$. So $\bigcap C \subseteq \mathbb{N} \times H$. Next let arbitrary $X \in C$. By definition $\langle 1, e \rangle \in X$, so $\langle 1, e \rangle \in \bigcap C$. Finally let n and y be arbitrary and suppose $\langle n, y \rangle \in \bigcap C$. We need to show that $\langle s(n), k(y) \rangle \in \bigcap C$. Let arbitrary $X \in C$. We have $\langle n, y \rangle \in X$. Since $X \in C$, by definition we must then have $\langle s(n), k(y) \rangle \in X$. We are done.

Next we note that $\forall X \in C (f \subseteq X)$. This is straightforward to prove. Finally we show

$$\begin{aligned} \forall n \forall y ([\langle n, y \rangle \in f \wedge \langle n, y \rangle \neq \langle 1, e \rangle] \\ \rightarrow \exists m \exists u (\langle m, u \rangle \in f \wedge \langle n, y \rangle = \langle s(m), k(u) \rangle)) \end{aligned}$$

□